# Learning manifolds with variational autoencoders

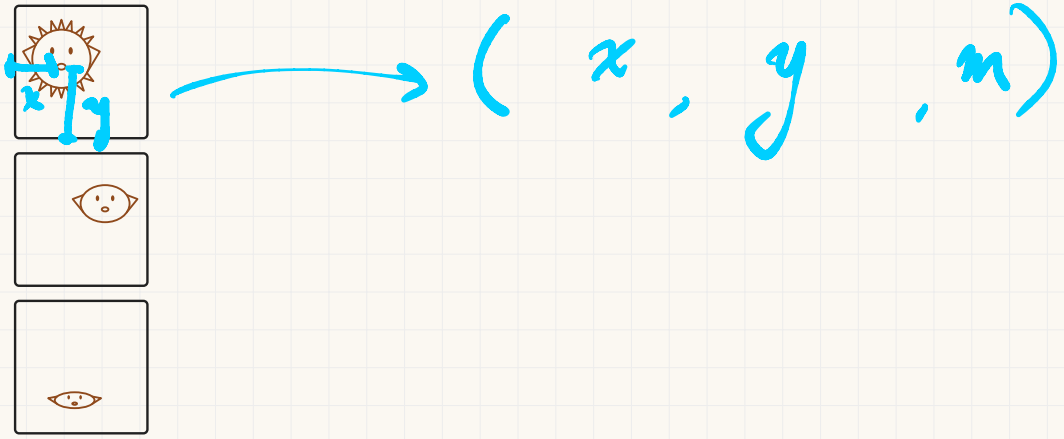
Weekend 4, Friday morning

Carlos Cotrini

*CAS Building ML/AI Applications, ETH Zurich*

*Friday 23 October 2026*

# Three numbers per picture



# Three numbers per picture



$\longrightarrow (x, y, m)$

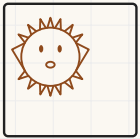
$x, y$: where the fish is

$m$: how puffed up it is



$m = 0$ $m = 0.5$ $m = 1$

# Three numbers per picture



$(x, y, m) = (0.35,\ 0.60,\ 1.00)$



$(x, y, m) = (0.68,\ 0.62,\ 0.50)$



$(x, y, m) = (0.45,\ 0.25,\ 0.00)$

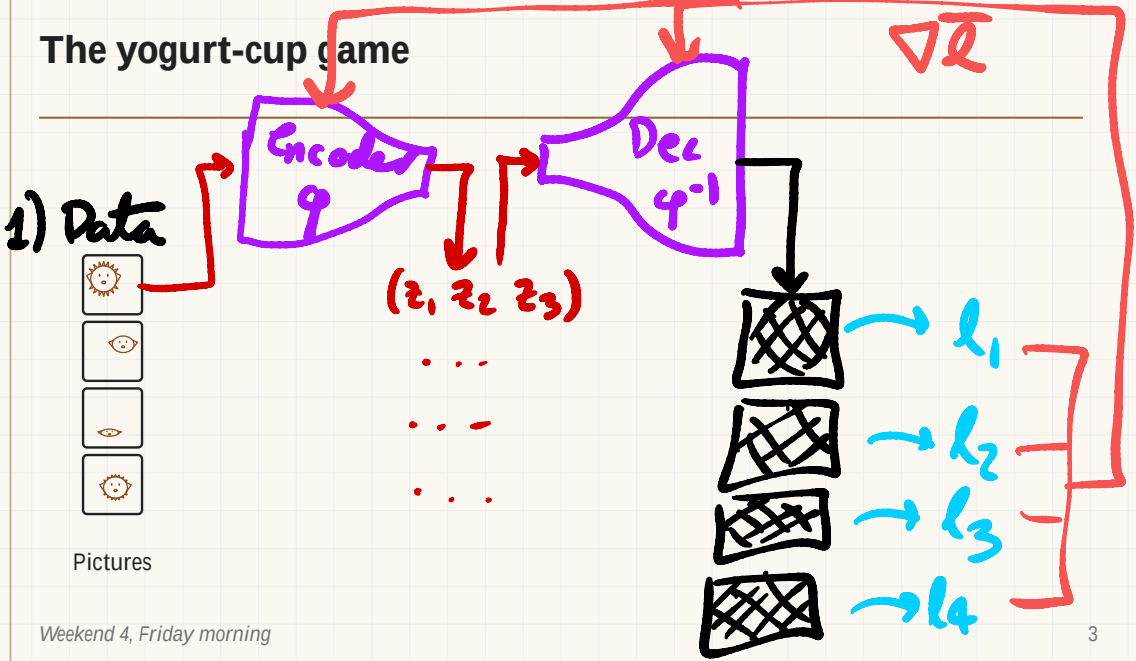
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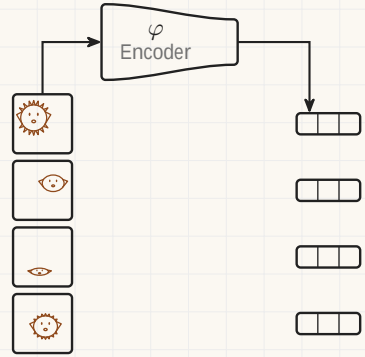
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# The yogurt-cup game




Pictures

# The yogurt-cup game

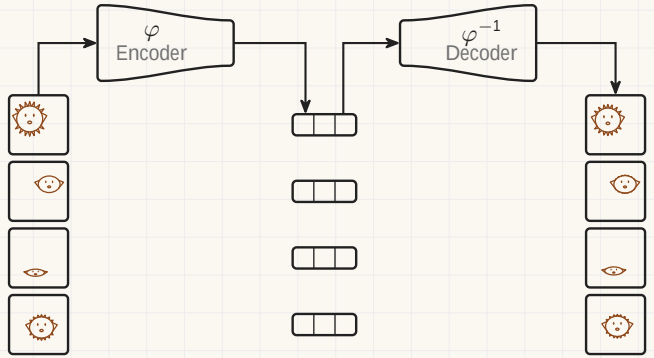



Pictures

Representations

Three numbers per picture

# The yogurt-cup game



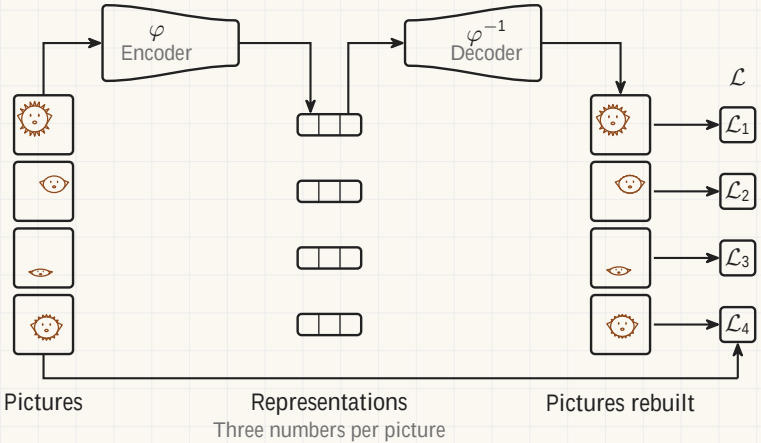

Pictures

Representations

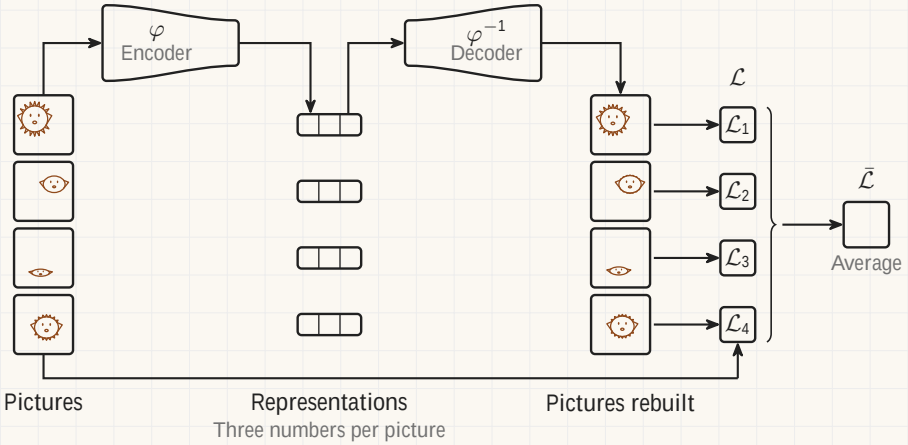
Three numbers per picture

Pictures rebuilt

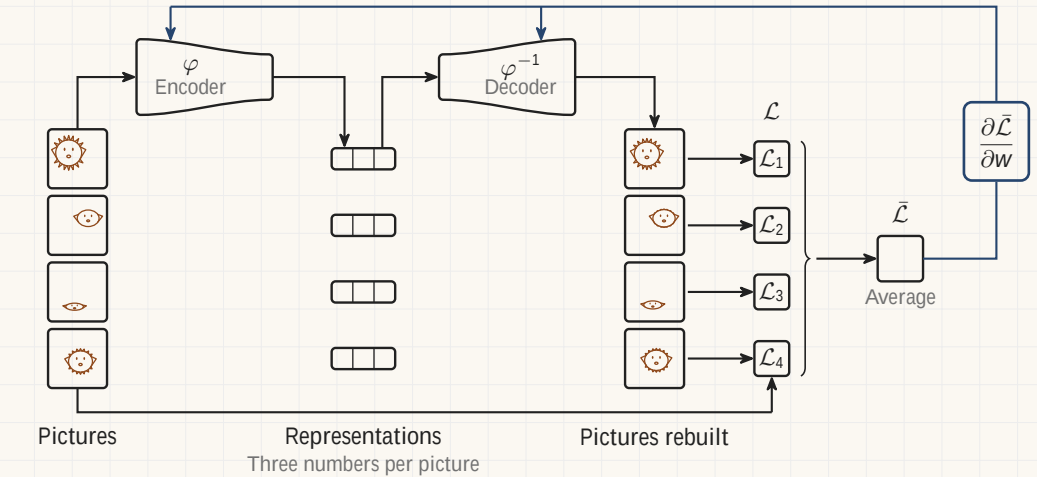
# The yogurt-cup game



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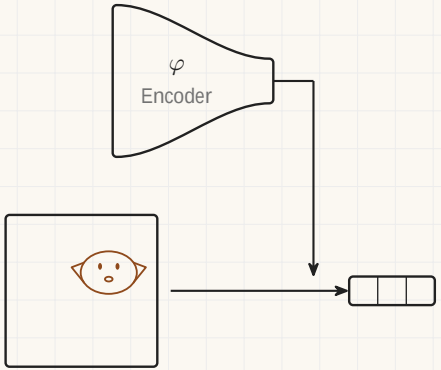


# Changing the representation

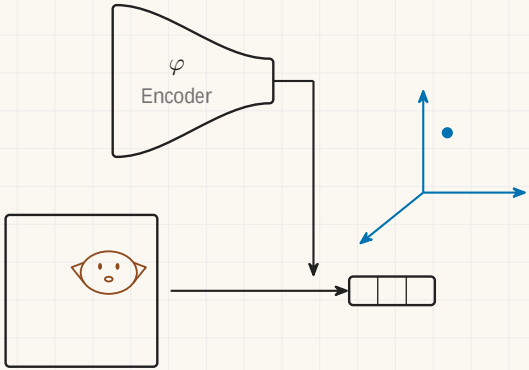


Picture

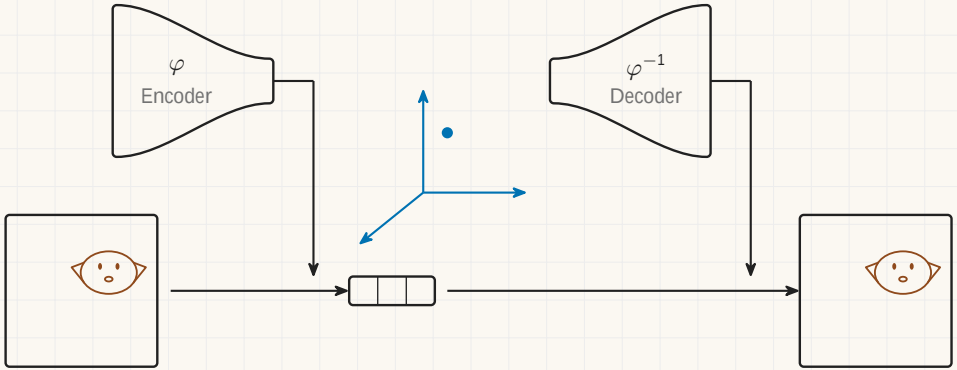
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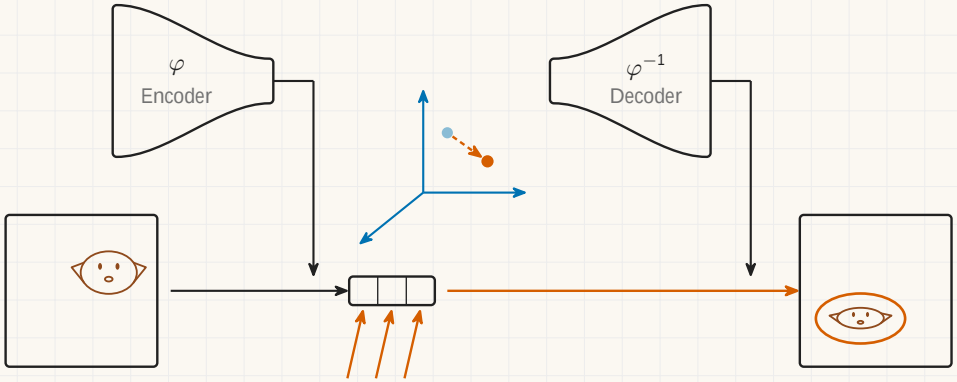
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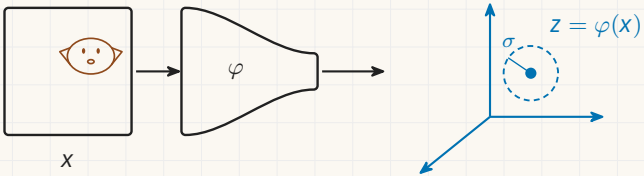
# The pu□er□sh notebook



1. 5000 made-up pu□er□sh pictures, each drawn from its three numbers $(x, y, m)$.
2. A plain autoencoder squeezes every picture into three numbers of its own, and back.
3. Your part: □ll in the three gaps marked ◎, then train it, about a minute and a half on Colab.
4. The last cell is the previous slide, live: pick a □sh, change its three numbers, see the new □sh.

`https://colab.research.google.com/github/eth-bmai-hs26/w4-cx-public/blob/main/pufferfish-autoencoder/pufferfish_autoencoder.ipynb`

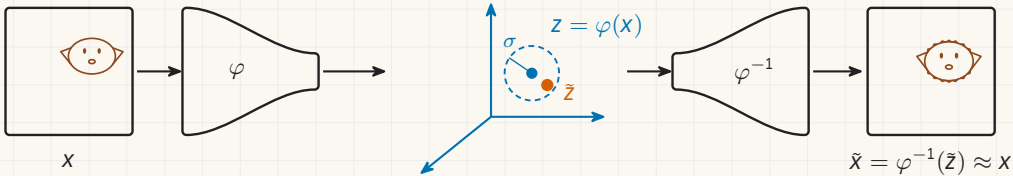
# Variational autoencoders




1. The encoder gives each picture a centre $z = \varphi(x)$ and a radius $\sigma$.

Kingma and Welling, Auto-Encoding Variational Bayes, ICLR 2014

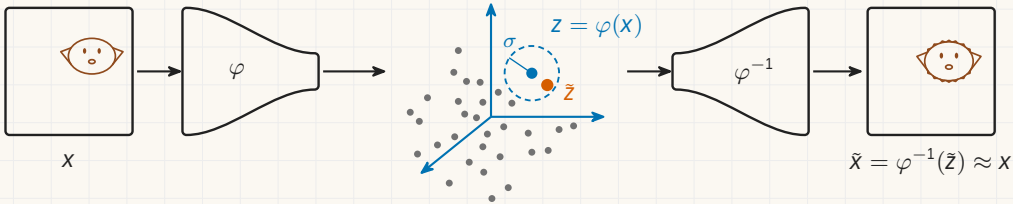
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2. The decoder gets a random $\tilde{z}$ within about $\sigma$ of $z$, and must still rebuild $x$.

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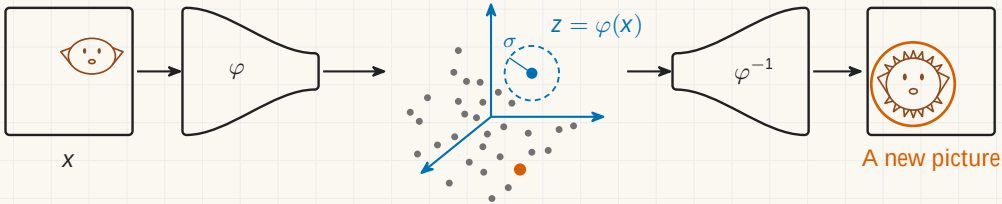
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3. A second term in the loss pulls the codes of all pictures into one round cloud.
4. To make a new picture, pick any point in the cloud and decode it.

Kingma and Welling, Auto-Encoding Variational Bayes, ICLR 2014